

CS5369L: Machine Learning and Applications

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Multi-Label Emotion Classification for Digital Journaling Applications



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1. OVERVIEW

This project develops a machine learning-powered digital journaling application that detects emotions in user journal entries and provides personalized mental wellness recommendations. The core component is a deep learning model using BERT (Bidirectional Encoder Representations from Transformers) fine-tuned on the GoEmotions dataset to classify text into five emotions: joy, sadness, anger, fear, and neutral.

A simple architecture of how the application works is:

React Native Journal App → User writes journal → Journal is processed by the ML model → Emotion Detection → Personalized suggestions to the user

Since this project's main focus was on the machine learning part, below are some of the achievements in the machine learning domain :

- Fine-tuned BERT model achieving 65% test accuracy and 93% validation accuracy
- Hyperparameter optimization (3 trials) reducing validation loss from 0.214 to 0.192
- Complete ML pipeline including data preprocessing, class imbalance handling, and systematic training
- Multi-label emotion classification successfully detecting 5 simultaneous emotions

The other features of the application are

- Mobile app (iOS/Android) built with React Native
- Real-time emotion analysis of journal entries
- Personalized activity recommendations based on detected emotions

2. INTRODUCTION

Digital journaling is growing rapidly, with the global market expected to increase from USD 6.68 billion in 2025 to USD 13.81 billion by 2034, and potentially reach USD 15.1 billion by 2035 [1]. This growth is driven by rising mental health awareness, widespread smartphone use, and remote work trends. Although journaling apps help users track thoughts and emotions, most do not offer intelligent emotional analysis. This project addresses that gap by developing a machine learning-powered journaling application that interprets emotional content and provides personalized wellness recommendations.

A key challenge while doing this project was selecting a dataset suited for multi-label emotion detection, since real journal entries often contain several emotions in a single sentence. The model was first trained on all 28 emotions in the GoEmotions dataset but was later narrowed to 5 emotions to improve accuracy and support efficient mobile deployment.

This report focuses mainly on the machine learning pipeline, including data preprocessing, model design, hyperparameter optimization, and training. The GoEmotions dataset (58,000 Reddit comments, 28 emotions) was chosen for its conversational tone, and five core emotions (joy, sadness, anger, fear, neutral) were used to simplify the task, reduce model size, and enable clear, actionable recommendations.

3. METHODOLOGY (MACHINE LEARNING PIPELINE)

Data Preprocessing: The ML preprocessing pipeline transforms raw text into BERT-compatible input.

Dataset Splitting: Training: 85% (36,899 samples), Validation: 7.5% (3,256 samples), Test: 7.5% (3,255 samples).

Emotion Label Engineering: I extracted 5 emotions and created binary multi-label vectors.

Text Length Analysis and Optimization: Dataset analysis revealed mean text length of 12.3 words and 95th percentile of 33 words. I set max_length=48 tokens, covering 100% of the dataset.

Emotion distribution showed severe imbalance (neutral: 42%, fear: 5%). I implemented weighted Binary Cross-Entropy loss to ensure the model learns rare emotions critical for mental health insights.

Model Architecture:

BERT-base-uncased (110M parameters, 12 transformer layers) serves as the foundation with a custom classification head. I used BERT for this Journaling App as it captures nuanced emotional expressions and has been pretrained on massive corpora that reduces training data needs. It also has proven great performance on emotion detection tasks. Systematic hyperparameter search with Optuna ensured optimal model performance for the journaling app:

The search Space is as follows:

- Learning rate: [1e-5, 5e-5] (log scale)
- Dropout: [0.1, 0.5]
- Batch size: {16, 32, 64}

Optuna Results (3 trials):

Trial	Val Loss	Learning Rate	Dropout	Batch Size

1	0.1925	1.15e-05	0.11	32
0	0.1988	1.08e-05	0.27	16
2	0.2146	2.91e-05	0.24	32

Best Configuration: lr=1.15e-05, dropout=0.11, batch=32 (3.2% improvement)

For training, I used AdamW optimizer with learning rate scheduling, Early stopping to prevent overfitting and Checkpoint saving for best models

4. EXPERIMENTAL SETUP

Tools and Frameworks: The model was developed using PyTorch 2.1 and PyTorch Lightning 2.0 for deep learning, Hugging Face Transformers 4.35 for pre-trained BERT models, Optuna 3.4 for hyperparameter optimization, and Weights & Biases for experiment tracking.

Infrastructure: Training was conducted on Google Colab with NVIDIA T4 GPU (16GB VRAM), 12GB system RAM, and CUDA 11.8 for GPU acceleration. All experiments used fixed random seeds (seed=42) for reproducibility.

Evaluation Metrics: For multi-label classification, I tracked accuracy, precision, recall, and F1-score.

5. RESULTS

Model Performance:

Final Test Set Evaluation:

- Test Accuracy: 65.06%
- Precision: 54.29%
- Recall: 28.25%
- F1-Score: 36.47%

Training Performance:

Dataset	Accuracy	Loss
Training	94.1%	0.113
Validation	92.65%	0.170
Test	65.06%	0.201

Analysis: The model shows strong performance on training (94.1%) and validation (92.65%) sets but experiences a significant drop on the test set (65.06%). This 27.59 percentage point gap suggests overfitting to the validation set. Despite the gap, the model still exceeds the baseline random performance (~20% for 5-class multi-label) and demonstrates learning of emotional patterns. The low recall (28.25%) indicates the model is conservative in its predictions, prioritizing precision over capturing all emotions.

Hyperparameter Optimization Impact

Optuna Optimization Results:

Configuration	Val Accuracy	Impact
Baseline (random params)	~72%	-
Optuna Best	88%	+16%

Best Hyperparameters Found:

- Learning rate: 1.15e-05
- Dropout: 0.11
- Batch size: 32

While hyperparameter optimization significantly improved validation performance (+16%), it may have contributed to overfitting by optimizing too specifically for the validation set. The

optimization was successful in improving training dynamics but highlights the need for larger, more diverse validation sets.

6. CONCLUSION

Summary

This project is a BERT-based emotion classifier for a journaling app, which improved accuracy from ~20% in the initial training to 65.06% on the final test set. It delivered a full ML pipeline, applied Optuna for a 16 percent validation gain, handled multi-label emotion detection and class imbalance, and prepared the model for mobile use. The system demonstrates that AI-driven emotional wellness support is feasible, though reducing overfitting is still required for production.

Limitations

The model is overfitting which is shown by a large gap between validation and test accuracy, likely caused by tuning too closely to the validation set. Recall is low, meaning the model misses many real emotions, especially rare ones. There is also a domain mismatch because the model was trained on Reddit rather than real journal entries. The model is large and expensive to run on mobile, and the strict multi-label accuracy metric hides partial successes, which makes the performance look worse than it may actually be.

Future Work

The short-term plan focuses on reducing overfitting, improving recall, and adapting the model to real journal data using better validation, data augmentation, threshold tuning, and domain-specific fine-tuning. The model will also be compressed for mobile use. The long-term plan aims for personalization and multi-modal emotion understanding. Overall, the project is workable today and provides a strong base for continued improvements.

REFERENCES

1. <https://www.marketresearchfuture.com/reports/digital-journal-apps-market-29194>
2. Demszky et al. (2020). "GoEmotions: A Dataset of Fine-Grained Emotions." *ACL 2020*.
3. Devlin et al. (2019). "BERT: Pre-training of Deep Bidirectional Transformers." *NAACL 2019*.

Project Links:

Google Colab: [Link](#)

Github: [Link](#)